DA2402 Data Curation and Visualisation Project

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Cleaning Log

#	Action and outcome
Header Reshaping	
1	Loaded raw file with no header, found a two row Urban/Rural/Total header structure (31 rows, 182 columns).
2	Dropped 18 fully empty trailing columns.
3	Forward filled the merged indicator names in row 0.
4	Combined each indicator name with its Urban, Rural or Total label into one flat column name. 164 columns remained.
Data Type Correction and Name Fixes	
5	Stripped Indian-numbering commas from the four money and population columns before casting to float.
6	Coerced all remaining columns to numeric. Result: 151 float64, 12 int64, 1 object column.
7	Corrected state name spelling: “Chattisgarh” to “Chhatisgarh”.
8	Corrected state name spelling: “Uttarkhand” to “Uttarakhand”.
9	Verified 29 unique, correctly spelled state names.
Feature Engineering	
10	Added Per_Capita_Health_Exp_Accounts (Accounts basis).
11	Added Per_Capita_Health_Exp_BE (Budget Estimates basis).
12	Added Per_Capita_Health_Exp_RE (Revised Estimates basis). Total columns now 167.
Duplicate Check	
13	Checked for fully duplicated rows. None found.
14	Checked for duplicate state names. None found.

#	Action and outcome
Outlier Detection	
15	Ran IQR outlier test on all numeric columns.
16	Ran modified Z-score (MAD based, threshold 3.5) outlier test.
17	Ran Grubbs' test (two sided, $\alpha = 0.05$) outlier test.
18	Reduced 166 numeric columns to 14 principal components (90% variance) for multivariate outlier detection.
19	Ran DBSCAN ($\epsilon = 12.0$, min_samples $= 3$) on the PCA reduced space. 6 states flagged as noise.
20	Ran Local Outlier Factor ($k = 3$) on the same space. 0 states flagged.
21	Cross checked flags across all five methods. All 6 DBSCAN flagged states also appeared in at least one univariate flag.
22	Decision: retained all data points. Flagged states reflect genuine small state variation, not errors.
Missing Value Analysis	
23	Counted missing values per column. 34 of 167 columns affected, out of 29 rows.
24	Reviewed co-missingness pattern. Urban columns largely missing together for states with little or no urban population.
25	Ran Little's global MCAR test. $p \approx 0.0000$, rejecting complete randomness, read with caution given 166 columns against 29 rows.
Imputation	
26	Decided against mean or median imputation, since it would erase state level variation.
27	Selected correlation guided KNN imputation as the single method for all missing columns.
28	For each missing column, identified its top 6 correlated numeric columns.
29	Ran KNNImputer ($k = 5$) on each column restricted to its own top correlated features, processed in ascending order of missing count.
30	Verified zero missing values remained after imputation.
Save	
31	Saved the cleaned dataset as budget_nfhs_combined_clean.csv.